

A COUNTEREXAMPLE TO THE CHUNG–GRAHAM–SPIRO GAP-SET CONJECTURE

MOHSEN ALIABADI

ABSTRACT. Chung, Graham, and Spiro introduced slow Fibonacci walks and used them to partition the integers $n \geq 2$ into two sequences, the down-integers and the up-integers. They studied the local spacing of these two sequences and conjectured that their ℓ -step gap sets agree for every $\ell \geq 1$. We show that the conjecture fails at $\ell = 4$ by proving

$$9 \in U_4 \setminus D_4.$$

1. INTRODUCTION

Let

$$f_1 = f_2 = 1, \quad f_{k+2} = f_{k+1} + f_k \quad (k \geq 1)$$

be the Fibonacci sequence, and let

$$\varphi = \frac{1 + \sqrt{5}}{2}.$$

Given positive integers a_1, a_2 , the corresponding *Fibonacci walk* is the sequence

$$w_1 = a_1, \quad w_2 = a_2, \quad w_{k+2} = w_{k+1} + w_k.$$

If $w_s = n$ for some s , then this walk reaches n . Among all Fibonacci walks that reach a fixed integer n , Chung, Graham, and Spiro [1] studied those that reach n as late as possible; these are the slow Fibonacci walks.

We shall use the following representation theorem from [1]. For every integer $n \geq 2$, there exist unique integers a, b, t such that

$$n = af_t + bf_{t-1}, \quad t \geq 2, \quad 1 \leq a \leq b \leq f_t.$$

We call this the *Chung–Graham–Spiro representation* of n , and we call t the *representation parameter* of n . A representation satisfying the displayed conditions will be called *valid*.

The slow walk reaching n has a next term. In the terminology of [1], an integer $n \geq 2$ is called a *down-integer* if this next term is $\lfloor \varphi n \rfloor$, and it is called an *up-integer* if this next term is $\lceil \varphi n \rceil$. The Chung–Graham–Spiro characterization says that n is a down-integer precisely when its representation parameter t is even, and that n is an up-integer precisely when t is odd. Thus the down-integers and the up-integers form a disjoint partition of the integers $n \geq 2$.

Let

$$D = \{d_1 < d_2 < d_3 < \cdots\}$$

be the increasing sequence of down-integers, and let

$$U = \{u_1 < u_2 < u_3 < \cdots\}$$

be the increasing sequence of up-integers. The first few values are

$$D = \{2, 5, 7, 9, 10, 12, 13, 15, 18, \dots\},$$

$$U = \{3, 4, 6, 8, 11, 14, 16, 17, \dots\}.$$

2020 *Mathematics Subject Classification*. 11B39, 11B37, 11B83, 05A15.

Key words and phrases. Fibonacci numbers, slow Fibonacci walks, down-integers, up-integers, gap sets, fractional parts.

These lists suggest that both sequences have very small local gaps between consecutive terms. Indeed, it was proved in [1] that the possible gaps between consecutive terms are the same for the two sequences, namely

$$D_1 = U_1 = \{1, 2, 3, 5\}.$$

Thus every two consecutive down-integers, and every two consecutive up-integers, differ by at most 5.

It is then natural to ask whether the same agreement persists for longer local spacings. For $\ell \geq 1$, define the ℓ -step gap sets

$$D_\ell = \{d_{k+\ell} - d_k : k \geq 1\}, \quad U_\ell = \{u_{k+\ell} - u_k : k \geq 1\}.$$

Thus D_ℓ records the possible spans of blocks of $\ell+1$ consecutive down-integers, and U_ℓ has the analogous meaning for up-integers. Chung, Graham, and Spiro also proved that

$$D_2 = U_2 = \{2, 3, 4, 5, 6, 8, 10\},$$

and conjectured that

$$D_\ell = U_\ell \quad \text{for every } \ell \geq 1.$$

The purpose of this note is to give a counterexample.

Theorem 1.1. *One has*

$$9 \in U_4 \setminus D_4.$$

In particular,

$$D_4 \neq U_4.$$

We first note that $9 \in U_4$. Using the representation criterion, the integers $2 \leq m \leq 17$ split as

$$D \cap [2, 17] = \{2, 5, 7, 9, 10, 12, 13, 15\},$$

$$U \cap [2, 17] = \{3, 4, 6, 8, 11, 14, 16, 17\}.$$

Hence

$$8, 11, 14, 16, 17$$

are five consecutive up-integers, and

$$17 - 8 = 9.$$

Therefore $9 \in U_4$. It remains to prove that $9 \notin D_4$.

For a real number x , let

$$\{x\} = x - \lfloor x \rfloor$$

denote its fractional part. For an integer $n \geq 1$, set

$$\delta_n = \{\varphi n\}.$$

Also define

$$\eta = \frac{1}{\sqrt{5}\varphi} = \frac{1}{\varphi+2}, \quad \rho = 1 - \eta.$$

We shall use the following consequence of Proposition 4.2 of [1].

Lemma 1.2. *For every integer $m \geq 2$,*

$$\delta_m < \eta \implies m \in D.$$

Consequently,

$$m \in U \implies \delta_m \geq \eta, \quad m \in D \implies \delta_m \leq \rho.$$

Proof. The first implication is Proposition 4.2 of [1]. Since each integer $m \geq 2$ lies in exactly one of D and U , its contrapositive gives

$$m \in U \implies \delta_m \geq \eta.$$

The same proposition also gives the upper-end implication

$$1 - \delta_m < \eta \implies m \in U.$$

Therefore, if $m \in D$, then $1 - \delta_m \geq \eta$, and hence

$$\delta_m \leq 1 - \eta = \rho.$$

□

For $0 \leq j \leq 9$, put

$$\alpha_j = \{j\varphi\}.$$

Then, for every integer $n \geq 1$,

$$\delta_{n+j} = \{\delta_n + \alpha_j\}.$$

Explicitly,

$$\begin{aligned}\alpha_0 &= 0, \\ \alpha_1 &= \varphi - 1, \\ \alpha_2 &= 2\varphi - 3, \\ \alpha_3 &= 3\varphi - 4, \\ \alpha_4 &= 4\varphi - 6, \\ \alpha_5 &= 5\varphi - 8, \\ \alpha_6 &= 6\varphi - 9, \\ \alpha_7 &= 7\varphi - 11, \\ \alpha_8 &= 8\varphi - 12, \\ \alpha_9 &= 9\varphi - 14.\end{aligned}$$

We shall also use two elementary Fibonacci identities. For even t ,

$$8 = f_{t-7}f_t - f_{t-6}f_{t-1},$$

and

$$5 = f_{t-5}f_{t-1} - f_{t-6}f_t.$$

These follow from d'Ocagne's identity

$$f_r f_{s+1} - f_{r+1} f_s = (-1)^s f_{r-s},$$

with the standard extension

$$f_{-m} = (-1)^{m+1} f_m.$$

For background on Fibonacci identities, see [2].

2. TWO SHIFT LEMMAS

The following two lemmas are the main structural ingredients in the proof.

Lemma 2.1 (Forward 8-shift). *Let $N \in D$, and write its Chung–Graham–Spiro representation as*

$$N = af_t + bf_{t-1}, \quad 1 \leq a \leq b \leq f_t.$$

Assume that t is even, $t \geq 8$, and

$$\delta_N \geq 15 - 9\varphi.$$

Then

$$N + 8 \in D.$$

Proof. Since $N \in D$ and t is even, Lemma 4.1 of [1] gives

$$\delta_N = \varphi^{-t}(\varphi b - a).$$

Thus

$$\varphi b - a \geq (15 - 9\varphi)\varphi^t.$$

We first prove that

$$b - a \geq f_{t-5}.$$

Suppose not. Since $b - a$ is an integer,

$$b - a \leq f_{t-5} - 1.$$

Using $b \leq f_t$, we get

$$\varphi b - a = \varphi^{-1}b + (b - a) \leq \varphi^{-1}f_t + f_{t-5} - 1.$$

If $t = 8$, then

$$\varphi^{-1}f_8 + f_3 - 1 = 21\varphi^{-1} + 1 = 21\varphi - 20,$$

whereas

$$(15 - 9\varphi)\varphi^8 = 9\varphi + 6.$$

The inequality

$$21\varphi - 20 < 9\varphi + 6$$

is equivalent to $12\varphi < 26$, which is true. Hence the desired contradiction holds when $t = 8$.

Now assume $t \geq 10$. We use the elementary estimate

$$f_s \leq \frac{23}{50}\varphi^s \quad (s \geq 4),$$

which follows directly from Binet's formula [2]. Since $t \geq 10$, both $t \geq 4$ and $t - 5 \geq 5$, so

$$f_t \leq \frac{23}{50}\varphi^t, \quad f_{t-5} \leq \frac{23}{50}\varphi^{t-5}.$$

Therefore

$$\varphi^{-1}f_t + f_{t-5} - 1 < \frac{23}{50}(\varphi^{-1} + \varphi^{-5})\varphi^t.$$

Using $\varphi^2 = \varphi + 1$, one obtains

$$15 - 9\varphi - \frac{23}{50}(\varphi^{-1} + \varphi^{-5}) = \frac{957 - 588\varphi}{50} > 0.$$

Hence

$$\varphi^{-1}f_t + f_{t-5} - 1 < (15 - 9\varphi)\varphi^t,$$

contradicting the lower bound for $\varphi b - a$. Therefore

$$b - a \geq f_{t-5}.$$

For even t ,

$$8 = f_{t-7}f_t - f_{t-6}f_{t-1}.$$

Thus

$$\begin{aligned} N + 8 &= af_t + bf_{t-1} + 8 \\ &= (a + f_{t-7})f_t + (b - f_{t-6})f_{t-1}. \end{aligned}$$

Set

$$a' = a + f_{t-7}, \quad b' = b - f_{t-6}.$$

Since

$$b - a \geq f_{t-5} = f_{t-6} + f_{t-7},$$

we have

$$b' - a' = b - a - f_{t-6} - f_{t-7} \geq 0.$$

Also $a' \geq 1$ and $b' \leq b \leq f_t$. Hence

$$1 \leq a' \leq b' \leq f_t.$$

Therefore $N + 8$ has a valid Chung–Graham–Spiro representation with the same even representation parameter t . Hence $N + 8 \in D$. $\square$

Lemma 2.2 (Backward 5-shift). *Let $N \in D$, and write its Chung–Graham–Spiro representation as*

$$N = af_t + bf_{t-1}, \quad 1 \leq a \leq b \leq f_t.$$

Assume that t is even, $t \geq 8$, and

$$\delta_N \geq 9\varphi - 14.$$

Then

$$N - 5 \in D.$$

Proof. Since $N \in D$ and t is even, Lemma 4.1 of [1] gives

$$\delta_N = \varphi^{-t}(\varphi b - a).$$

Thus

$$\varphi b - a \geq (9\varphi - 14)\varphi^t.$$

We first prove that

$$b - a \geq f_{t-4}.$$

Suppose not. Since $b - a$ is an integer,

$$b - a \leq f_{t-4} - 1.$$

Using $b \leq f_t$, we obtain

$$\varphi b - a = \varphi^{-1}b + (b - a) \leq \varphi^{-1}f_t + f_{t-4} - 1.$$

Since $t \geq 8$, both $t \geq 4$ and $t - 4 \geq 4$. Hence

$$f_t \leq \frac{23}{50}\varphi^t, \quad f_{t-4} \leq \frac{23}{50}\varphi^{t-4}.$$

Therefore

$$\varphi^{-1}f_t + f_{t-4} - 1 < \frac{23}{50}(\varphi^{-1} + \varphi^{-4})\varphi^t.$$

Using $\varphi^2 = \varphi + 1$, one obtains

$$9\varphi - 14 - \frac{23}{50}(\varphi^{-1} + \varphi^{-4}) = \frac{248\varphi - 396}{25} > 0.$$

Thus

$$\varphi^{-1}f_t + f_{t-4} - 1 < (9\varphi - 14)\varphi^t,$$

contradicting the lower bound for $\varphi b - a$. Therefore

$$b - a \geq f_{t-4}.$$

For even t ,

$$5 = f_{t-5}f_{t-1} - f_{t-6}f_t.$$

Hence

$$\begin{aligned} N - 5 &= af_t + bf_{t-1} - 5 \\ &= (a + f_{t-6})f_t + (b - f_{t-5})f_{t-1}. \end{aligned}$$

Set

$$a' = a + f_{t-6}, \quad b' = b - f_{t-5}.$$

Since

$$b - a \geq f_{t-4} = f_{t-5} + f_{t-6},$$

we have

$$b' - a' = b - a - f_{t-5} - f_{t-6} \geq 0.$$

Also $a' \geq 1$ and $b' \leq b \leq f_t$. Hence

$$1 \leq a' \leq b' \leq f_t.$$

Therefore $N - 5$ has a valid Chung–Graham–Spiro representation with the same even representation parameter t . Hence $N - 5 \in D$. $\square$

3. PROOF OF THE COUNTEREXAMPLE

We first dispose of a finite initial range.

Lemma 3.1. *There are no five consecutive down-integers whose first-to-fifth difference is 9 and such that at least one of the five has representation parameter $t < 8$.*

Proof. By five consecutive down-integers, we mean five terms

$$d_k, d_{k+1}, d_{k+2}, d_{k+3}, d_{k+4}$$

of the increasing sequence D .

If $M \in D$ has even representation parameter $t < 8$, then

$$t \in \{2, 4, 6\}.$$

Writing

$$M = af_t + bf_{t-1}, \quad 1 \leq a \leq b \leq f_t,$$

we get

$$M \leq f_t^2 + f_t f_{t-1} = f_t f_{t+1} \leq f_6 f_7 = 104.$$

Thus any interval $[n, n+9]$ containing such an M is contained in $[2, 113]$.

The finite enumeration is reproduced in the appendix. It implements the Chung–Graham–Spiro representation directly and gives

$$\begin{aligned} D \cap [2, 113] = \{ & 2, 5, 7, 9, 10, 12, 13, 15, 18, 23, 26, 28, 31, 33, 34, 36, 38, 39, 41, 43, \\ & 44, 46, 47, 48, 49, 51, 52, 54, 56, 57, 59, 60, 62, 64, 65, 67, 68, 70, \\ & 72, 73, 75, 78, 80, 81, 83, 86, 88, 89, 91, 94, 96, 99, 102, 104, 107, 112 \}. \end{aligned}$$

Checking the consecutive five-term blocks in this displayed list shows that no five consecutive elements of $D \cap [2, 113]$ have first-to-fifth difference 9. $\square$

We now prove that $9 \notin D_4$. Suppose, for contradiction, that $9 \in D_4$. Then there exist five consecutive down-integers

$$n = d_k, \quad d_{k+1}, \quad d_{k+2}, \quad d_{k+3}, \quad d_{k+4} = n + 9.$$

Let

$$g_i = d_{k+i} - d_{k+i-1} \quad (1 \leq i \leq 4).$$

Since $D_1 = \{1, 2, 3, 5\}$,

$$g_i \in \{1, 2, 3, 5\}.$$

Since $D_2 = \{2, 3, 4, 5, 6, 8, 10\}$,

$$g_i + g_{i+1} \in \{2, 3, 4, 5, 6, 8, 10\} \quad (1 \leq i \leq 3).$$

Finally,

$$g_1 + g_2 + g_3 + g_4 = 9.$$

A direct enumeration gives exactly the following twenty-two possible patterns:

$$\begin{aligned} & (1, 2, 1, 5), (1, 2, 3, 3), (1, 3, 2, 3), (1, 3, 3, 2), (1, 5, 1, 2), \\ & (2, 1, 1, 5), (2, 1, 3, 3), (2, 1, 5, 1), (2, 2, 2, 3), (2, 2, 3, 2), \\ & (2, 3, 1, 3), (2, 3, 2, 2), (2, 3, 3, 1), \\ & (3, 1, 2, 3), (3, 1, 3, 2), (3, 2, 1, 3), (3, 2, 2, 2), (3, 2, 3, 1), \\ & (3, 3, 1, 2), (3, 3, 2, 1), (5, 1, 1, 2), (5, 1, 2, 1). \end{aligned}$$

For a pattern $P = (g_1, g_2, g_3, g_4)$, define its *position set* $S(P)$ by

$$S(P) = \{0, g_1, g_1 + g_2, g_1 + g_2 + g_3, 9\}.$$

Thus $S(P)$ is the set of positions in the interval $[n, n+9]$ occupied by the five down-integers. Put

$$x = \delta_n.$$

If $j \in S(P)$, then $n + j \in D$, so Lemma 1.2 gives

$$\{x + \alpha_j\} \leq \rho.$$

If $j \notin S(P)$, then $n + j \notin D$. Since D and U partition the integers $m \geq 2$, this means $n + j \in U$, and Lemma 1.2 gives

$$\{x + \alpha_j\} \geq \eta.$$

The exact interval computations in Section 4 show that twelve of the twenty-two patterns have no feasible value of x , namely

$$(1, 5, 1, 2), (2, 1, 1, 5), (2, 1, 3, 3), (2, 1, 5, 1), (2, 2, 2, 3), (2, 2, 3, 2), \\ (3, 2, 1, 3), (3, 2, 2, 2), (3, 3, 1, 2), (3, 3, 2, 1), (5, 1, 1, 2), (5, 1, 2, 1).$$

For the remaining ten patterns, the same exact computation gives the feasible intervals

P	feasible interval for $x = \delta_n$
$(1, 2, 1, 5)$	$[\eta + 1 - \alpha_6, 1 - \alpha_7]$
$(1, 2, 3, 3)$	$[1 - \alpha_9, 1 - \alpha_4]$
$(1, 3, 2, 3)$	$[1 - \alpha_4, 1 - \alpha_7]$
$(1, 3, 3, 2)$	$[1 - \alpha_7, \rho]$
$(2, 3, 1, 3)$	$[0, \rho - \alpha_6]$
$(2, 3, 2, 2)$	$[0, 1 - \alpha_8]$
$(2, 3, 3, 1)$	$[1 - \alpha_8, 1 - \alpha_3]$
$(3, 1, 2, 3)$	$[\eta + 1 - \alpha_1, 1 - \alpha_7]$
$(3, 1, 3, 2)$	$[1 - \alpha_7, \rho]$
$(3, 2, 3, 1)$	$[1 - \alpha_3, \rho - \alpha_9]$

Consider first the six feasible patterns

$$(1, 2, 1, 5), (1, 2, 3, 3), (1, 3, 2, 3), (1, 3, 3, 2), \\ (3, 1, 2, 3), (3, 1, 3, 2).$$

For each of these patterns, the displayed feasible interval implies

$$\delta_n \geq 15 - 9\varphi.$$

Moreover, each of these six patterns has

$$n + 8 \in U.$$

By Lemma 3.1, the down-integer n has even representation parameter $t \geq 8$. Hence Lemma 2.1 gives

$$n + 8 \in D,$$

contradicting $n + 8 \in U$.

It remains to consider the four feasible patterns

$$(2, 3, 1, 3), (2, 3, 2, 2), (2, 3, 3, 1), (3, 2, 3, 1).$$

Set

$$N = n + 9.$$

Then $N \in D$. For each of these four patterns, the displayed feasible interval implies

$$x \leq \rho - \alpha_9.$$

Thus

$$x + \alpha_9 \leq \rho < 1,$$

and so

$$\delta_N = \delta_{n+9} = \{x + \alpha_9\} = x + \alpha_9.$$

Since $x \geq 0$, this gives

$$\delta_N \geq \alpha_9 = 9\varphi - 14.$$

Each of these four patterns also has

$$n + 4 = N - 5 \in U.$$

By Lemma 3.1, the down-integer N has even representation parameter $t \geq 8$. Hence Lemma 2.2 gives

$$N - 5 = n + 4 \in D,$$

contradicting $n + 4 \in U$.

Thus every possible pattern is impossible. Therefore

$$9 \notin D_4.$$

Since $9 \in U_4$, we conclude that

$$9 \in U_4 \setminus D_4.$$

This proves Theorem 1.1.

4. EXACT INTERVAL CERTIFICATES

This section records the exact interval computations used in the proof. All computations take place in the quadratic field $\mathbb{Q}(\varphi)$ and use only the relation

$$\varphi^2 = \varphi + 1.$$

For $0 \leq j \leq 9$, define

$$I_D(j) = \{x \in [0, 1) : \{x + \alpha_j\} \leq \rho\},$$

and

$$I_U(j) = \{x \in [0, 1) : \{x + \alpha_j\} \geq \eta\}.$$

For a pattern P , the feasible set for $x = \delta_n$ is obtained by intersecting $I_D(j)$ over all $j \in S(P)$ and $I_U(j)$ over all $j \notin S(P)$.

For $0 \leq \alpha < 1$, the condition

$$\{x + \alpha\} \leq \rho$$

is equivalent to

$$x \in ([0, \rho - \alpha] \cap [0, 1)) \cup ([1 - \alpha, 1 + \rho - \alpha] \cap [0, 1)).$$

Similarly,

$$\{x + \alpha\} \geq \eta$$

is equivalent to

$$x \in ([\eta - \alpha, 1 - \alpha] \cap [0, 1)) \cup ([1 + \eta - \alpha, 1] \cap [0, 1)).$$

Here endpoints equal to 1 are harmless, since throughout $x \in [0, 1)$.

For each pattern P , the intervals below are obtained by intersecting the five $I_D(j)$ -conditions for $j \in S(P)$ and the five $I_U(j)$ -conditions for $j \notin S(P)$. Substituting $\alpha = \alpha_j$, using

$$\eta = \frac{1}{\varphi + 2}, \quad \rho = 1 - \eta,$$

and reducing by $\varphi^2 = \varphi + 1$ gives the following certificates.

The next table proves that twelve patterns have empty feasible set. In each row, the displayed interval is forced by some of the necessary conditions for the pattern, while the final listed necessary condition is disjoint from that interval. Hence the full feasible intersection is empty.

P	forced interval	additional necessary condition
$(1, 5, 1, 2)$	$[\frac{28}{5} - \frac{16}{5}\varphi, 7 - 4\varphi]$	$I_D(7) = [0, \frac{57}{5} - \frac{34}{5}\varphi] \cup [12 - 7\varphi, 1]$
$(2, 1, 1, 5)$	$[\frac{43}{5} - \frac{26}{5}\varphi, 2 - \varphi]$	$I_D(9) = [0, \frac{72}{5} - \frac{44}{5}\varphi] \cup [15 - 9\varphi, 1]$
$(2, 1, 3, 3)$	$[\frac{43}{5} - \frac{26}{5}\varphi, 2 - \varphi]$	$I_D(9) = [0, \frac{72}{5} - \frac{44}{5}\varphi] \cup [15 - 9\varphi, 1]$
$(2, 1, 5, 1)$	$[\frac{43}{5} - \frac{26}{5}\varphi, 2 - \varphi]$	$I_D(9) = [0, \frac{72}{5} - \frac{44}{5}\varphi] \cup [15 - 9\varphi, 1]$
$(2, 2, 2, 3)$	$[0, 5 - 3\varphi]$	$I_U(5) = [\frac{43}{5} - \frac{26}{5}\varphi, 9 - 5\varphi]$
$(2, 2, 3, 2)$	$[0, 5 - 3\varphi]$	$I_U(5) = [\frac{43}{5} - \frac{26}{5}\varphi, 9 - 5\varphi]$
$(3, 2, 1, 3)$	$[10 - 6\varphi, 2 - \varphi]$	$I_D(9) = [0, \frac{72}{5} - \frac{44}{5}\varphi] \cup [15 - 9\varphi, 1]$
$(3, 2, 2, 2)$	$[5 - 3\varphi, 10 - 6\varphi]$	$I_U(8) = [0, 13 - 8\varphi] \cup [\frac{68}{5} - \frac{41}{5}\varphi, 1]$
$(3, 3, 1, 2)$	$[\frac{43}{5} - \frac{26}{5}\varphi, 2 - \varphi]$	$I_D(9) = [0, \frac{72}{5} - \frac{44}{5}\varphi] \cup [15 - 9\varphi, 1]$
$(3, 3, 2, 1)$	$[\frac{43}{5} - \frac{26}{5}\varphi, 2 - \varphi]$	$I_D(9) = [0, \frac{72}{5} - \frac{44}{5}\varphi] \cup [15 - 9\varphi, 1]$
$(5, 1, 1, 2)$	$[\frac{18}{5} - \frac{11}{5}\varphi, 5 - 3\varphi]$	$I_D(6) = [0, \frac{47}{5} - \frac{29}{5}\varphi] \cup [10 - 6\varphi, 1]$
$(5, 1, 2, 1)$	$[\frac{18}{5} - \frac{11}{5}\varphi, 5 - 3\varphi]$	$I_D(6) = [0, \frac{47}{5} - \frac{29}{5}\varphi] \cup [10 - 6\varphi, 1]$

For example, in the first row the forced interval is

$$\left[\frac{28}{5} - \frac{16}{5}\varphi, 7 - 4\varphi \right].$$

The additional condition $I_D(7)$ requires

$$x \leq \frac{57}{5} - \frac{34}{5}\varphi \quad \text{or} \quad x \geq 12 - 7\varphi.$$

But

$$\frac{57}{5} - \frac{34}{5}\varphi < \frac{28}{5} - \frac{16}{5}\varphi \leq x \leq 7 - 4\varphi < 12 - 7\varphi.$$

Thus the intersection is empty. The other rows are verified in exactly the same way.

For the ten nonempty cases, the same intersection procedure gives the following exact feasible intervals:

P	feasible interval
$(1, 2, 1, 5)$	$[\eta + 1 - \alpha_6, 1 - \alpha_7]$
$(1, 2, 3, 3)$	$[1 - \alpha_9, 1 - \alpha_4]$
$(1, 3, 2, 3)$	$[1 - \alpha_4, 1 - \alpha_7]$
$(1, 3, 3, 2)$	$[1 - \alpha_7, \rho]$
$(2, 3, 1, 3)$	$[0, \rho - \alpha_6]$
$(2, 3, 2, 2)$	$[0, 1 - \alpha_8]$
$(2, 3, 3, 1)$	$[1 - \alpha_8, 1 - \alpha_3]$
$(3, 1, 2, 3)$	$[\eta + 1 - \alpha_1, 1 - \alpha_7]$
$(3, 1, 3, 2)$	$[1 - \alpha_7, \rho]$
$(3, 2, 3, 1)$	$[1 - \alpha_3, \rho - \alpha_9]$

These are precisely the intervals used in the proof of Theorem 1.1.

5. DISCUSSION AND OPEN QUESTIONS

Note that the case $\ell = 3$ is not settled by the present argument. Our numerical computations have not revealed any discrepancy between D_3 and U_3 , providing some computational evidence that

$$D_3 = U_3.$$

It would be interesting to determine whether this equality indeed holds. More generally, for which values of ℓ do we have

$$D_\ell = U_\ell?$$

In particular, does the failure exhibited here at $\ell = 4$ persist for all, or infinitely many, larger values of ℓ ?

APPENDIX: FINITE VERIFICATION

The following Sage code reproduces the finite enumeration used in Lemma 3.1. It implements the Chung–Graham–Spiro representation directly and checks that no five consecutive down-integers in the required finite range have span 9.

```
def fibs_up_to(N):
    f = [0, 1, 1]
    while f[-1] <= N:
        f.append(f[-1] + f[-2])
    return f

def cgs_representation(m):
    f = fibs_up_to(m)
    for t in range(2, len(f)):
        if f[t] + f[t-1] > m:
            break
        for a in range(1, f[t] + 1):
            for b in range(a, f[t] + 1):
                if a*f[t] + b*f[t-1] == m:
                    return (a, b, t)
    raise ValueError("No representation found")

D = []

for m in range(2, 114):
    a, b, t = cgs_representation(m)
    if t % 2 == 0:
        D.append(m)

print("D cap [2,113] =")
print(D)

bad_blocks = []
for i in range(len(D) - 4):
    block = D[i:i+5]
    if block[-1] - block[0] == 9:
        bad_blocks.append(block)

print("Five-term down-blocks with span 9:")
print(bad_blocks)
```

The output is the displayed list of $D \cap [2, 113]$ in Lemma 3.1, followed by the empty list of five-term down-blocks with span 9.

Conflict of interest. The author declares that there is no conflict of interest.

Data availability. No datasets were generated or analyzed during the present work.

REFERENCES

- [1] F. Chung, R. L. Graham, and S. Spiro, Slow Fibonacci walks, *J. Number Theory* **210** (2020), 142–170.

- [2] T. Koshy, *Fibonacci and Lucas Numbers with Applications*, Wiley-Interscience, New York, 2001.

DEPARTMENT OF MATHEMATICS, CLAYTON STATE UNIVERSITY, MORROW, GA, USA
Email address: `mohsen.aliabadi@clayton.edu`